

New Solutions in the Last One Year for M2M-IoT Products

Company	Solution and type	Why you should pick this for your next IoT product?
Texas Instruments	 SimpleLink Wi-Fi CC3100 Internet-on-a-chip wireless network processor Chipset solution	- On-chip web server and embedded TCP/IP stack - Integrates all protocols for Wi-Fi and the Internet 64-pin 9x9mm QFN package - Can connect to any 8-, 16-, or 32-bit MCU over SPI or UART interface - UART or SPI driver minimises host-memory footprint as low as 7kB of code and 700B of RAM memory for a TCP client application
Texas Instruments	 CC3200 SimpleLink Wi-Fi and Internet-of-Things Chipset solution	- Wireless MCU that integrates ARM Cortex-M4 MCU running at 80MHz - Peripheral set includes parallel camera, I2S, SD/MMC, UART, SPI, I2C and four-channel ADC - Includes embedded TCP/IP and TLS/SSL stacks, HTTP server and multiple Internet protocols
Redpine Signals	 Lite-FS RS9110 Chipset solution	- Single antenna IEEE 802.11n with STBC support - Host interfaces include SDIO v1.2/2.0, SPI, memory-mapped interface (EBI) and other interfaces are I2C, SPI, GPIO and UART - WEP, TKIP and AES hardware engines for line-speed security support
Redpine Signals	 M2MCombo RS9113 Chipset solution	- On-chip processing and memory enables the option to offload Wi-Fi security supplicant - TCP/IP stack enabling zero-host load - Single-spatial stream, dual-band 802.11n + BT4.0 (Bluetooth LE dual-mode) + ZigBee (802.15.4-2006) - Host interfaces supported are USB 2.0, SDIO, SPI and UART
Sequans Communication	 Colibri LTE platform Chipset solution	- For adding LTE connectivity to M2M and IoT modules - FDD and TDD, up to 20MHz LTE channels - Supports VoLTE, location based services and Wi-Fi SoftAP - Host environments include Android, ChromeOS, Linux, Windows and MAC OS
Gemalto	 Clintier concept board Development board solution	- Development kit that leverages EHS6 Java module, Clintier solutions and SensorLogic application enablement platform for design of M2M solutions - Can be connected to Arduino-style sensor boards and includes a Clintier M2M module for global 2G and 3G cellular connectivity

wide adaptability, it needs to support all protocols, or the designer needs to segment devices and companies based on the kind of standards their machines support, as each protocol requires specific investment and knowledge-base that the designer needs.”

Long-distance communication. Long-distance communication is not

just about getting data to the server, it is about the whole stack—how to get the data to the server, how the server interprets and makes sense of it, informs Singh. Again, there are a lot of protocols, and each of these has its individual stack. Let us take a few examples.

Singh says, “The most common and easy-to-understand is the Trans-

mission Control Protocol/Internet Protocol (TCP/IP) based protocol. You could send the data out from your local gateway to a server via IP version 6 (IPv6). One big problem with TCP/IP is that it is very expensive with regards to power, and the overhead is also very high, especially if you have hundreds or thousands of devices sending data.”

IBM, a few years ago, came out with a protocol named Message Queuing Telemetry Transport (MQTT), which, Singh notes, is a short messaging protocol, where the device or gateway should broadcast the message over a hub. Individual servers or applications that need access to the data can subscribe to it. He adds, “There are many other proprietary protocols that companies like Schneider or GE have built on data-collation. So, there is a big problem of fragmentation and how to build the application in a way that you can deploy multiple vendor machines.”

You can add LTE connectivity to your designs

To make M2M- and IoT-enabled devices, there is a need for a simple Long-Term Evolution (LTE) platform, as industry experts believe LTE-enabled IoT is an upcoming trend. As LTE gains momentum around the world, the market for single-mode LTE module solution provides a huge opportunity, as modules could be a worthy option for many IoT device makers.

Electronics manufacturing companies have started incorporating LTE chipsets to bring to market a module optimised for the design of devices for the IoT. Such modules aim to target the needs of M2M and IoT device makers for a straightforward, all-in-one, single-mode LTE chip solution for the design of reasonably-priced devices for numerous IoT applications. It is slowly growing as more and more economies are changing to LTE infrastructure these days. Faster data rates and high bandwidth demands are also fuelling these requirements.